

Strings

LEVEL 5: Pattern Matching & Searching

Program 69: Linear Search for Substring Write a program that finds first occurrence of substring in main string. Display starting position.

Program 70: Count Occurrences of Substring Write a program that counts how many times a substring appears in the main string.

Program 71: Find All Occurrences of Substring Write a program that displays all starting positions where substring appears.

Program 72: Check if String Starts With Prefix Write a program that checks if string starts with a given prefix.

Program 73: Check if String Ends With Suffix Write a program that checks if string ends with a given suffix.

Program 74: Find Longest Word in String Write a program that finds and displays the longest word.

Program 75: Find Shortest Word in String Write a program that finds and displays the shortest word.

Program 76: Count Words Starting with Vowel Write a program that counts how many words start with a vowel.

Program 77: Count Words Ending with Vowel Write a program that counts how many words end with a vowel.

Program 78: Find Words of Specific Length Write a program that displays all words having exactly N characters.

Program 79: Check if Anagram Write a program that checks if two strings are anagrams (same characters, different order). Example: "listen" and "silent".

Program 80: Check if Rotation of Another String Write a program that checks if string A is rotation of string B. Example: "ABCD" is rotation of "CDAB".

LEVEL 6: Number-String Conversions

Program 81: String to Integer Conversion Write a program that converts a numeric string to integer. Example: "1234" → 1234.

Program 82: Integer to String Conversion Write a program that converts an integer to string representation.

Program 83: Count Numbers in String Write a program that counts how many numeric words or numbers are in the string.

Program 84: Extract All Digits from String Write a program that extracts all digits and forms a number. Example: "abc123def456" → 123456.

Program 85: Sum of All Digits in String Write a program that calculates sum of all digit characters. Example: "a1b2c3" → $1+2+3 = 6$.

Program 86: Extract All Numbers from String Write a program that extracts all complete numbers. Example: "I have 12 apples and 34 oranges" → 12, 34.

Program 87: Replace Numbers with Words Write a program that replaces digit characters with their word form. Example: "I am 21" → "I am two one".

Program 88: Check if String Represents Valid Number Write a program that checks if string can be converted to a valid number.

Program 89: Count Alphabets vs Digits Ratio Write a program that calculates ratio of alphabets to digits in the string.

Program 90: Remove All Numbers from String Write a program that removes all numeric characters from the string.

LEVEL 7: Advanced String Problems

Program 91: Check if Pangram Write a program that checks if string contains all 26 letters of alphabet. Example: "The quick brown fox jumps over the lazy dog".

Program 92: Check if Isogram Write a program that checks if no letter appears more than once (isogram). Example: "algorithm" → Yes.

Program 93: Check if Heterogram Write a program that checks if each character appears exactly once.

Program 94: Longest Palindromic Substring Write a program that finds the longest substring which is a palindrome.

Program 95: Count Palindromic Substrings Write a program that counts how many substrings are palindromes.

Program 96: Remove Duplicate Characters (Keep First) Write a program that removes duplicate characters keeping first occurrence. Example: "programming" → "progamin".

Program 97: Remove Duplicate Words Write a program that removes duplicate words from the string.

Program 98: Find Common Characters in Two Strings Write a program that finds characters present in both strings.

Program 99: Print All Permutations of String Write a program that generates and displays all permutations of a given string.

Program 100: Count Permutations of String Write a program that counts total number of unique permutations possible.

Program 101: Longest Common Substring Write a program that finds longest common substring between two strings.

Program 102: Longest Common Prefix Write a program that finds longest common prefix among array of strings.

Program 103: Check if Subsequence Write a program that checks if string A is subsequence of string B (characters appear in same order).

Program 104: Count Vowel Substrings Write a program that counts substrings containing only vowels.

Program 105: Toggle Vowels and Consonants Write a program that swaps vowels with consonants based on position.

Program 106: Caesar Cipher Encryption Write a program that encrypts string using Caesar cipher (shift each letter by K positions).

Program 107: Caesar Cipher Decryption Write a program that decrypts Caesar cipher encrypted text.

Program 108: Run Length Encoding Write a program that compresses string using run-length encoding. Example: "aaabbcccc" → "a3b2c4".

Program 109: Run Length Decoding Write a program that decompresses run-length encoded string.

Program 110: String Compression (Unique Characters Count) Write a program that compresses by writing character followed by count. Example: "aabbcc" → "a2b2c2".

LEVEL 8: String Sorting & Rearrangement

Program 111: Sort Words Alphabetically Write a program that sorts all words in string in alphabetical order.

Program 112: Sort Words by Length Write a program that sorts words by their length (shortest to longest).

Program 113: Sort Characters by Frequency Write a program that rearranges string with most frequent characters first.

Program 114: Rearrange Vowels and Consonants Write a program that rearranges with all vowels first, then consonants.

Program 115: Group Anagrams Together Write a program that groups all anagram words together from array of words.

Program 116: Find Lexicographically Smallest String Write a program that rearranges characters to form lexicographically smallest string.

Program 117: Find Lexicographically Largest String Write a program that rearranges characters to form lexicographically largest string.

Program 118: Next Lexicographic Permutation Write a program that finds next lexicographic permutation of string.

LEVEL 9: Advanced Pattern & Logic

Program 119: Validate Email Address Write a program that checks if string is a valid email format (contains @, domain, etc.).

Program 120: Validate Phone Number Write a program that checks if string is valid phone number format.

Program 121: Validate Password Strength Write a program that checks password strength (length, uppercase, lowercase, digit, special char).

Program 122: Check Balanced Parentheses Write a program that checks if parentheses (), {}, [] are balanced in the string.

Program 123: Reverse Words Preserving Spaces Write a program that reverses string but keeps space positions fixed.

Program 124: Find First Unique Word Write a program that finds first word that appears only once in the string.

Program 125: Longest Word Without Repeating Characters Write a program that finds longest word with all unique characters.

Program 126: Count Sentences in String Write a program that counts sentences (separated by . ! ?).

Program 127: Extract Acronym from String Write a program that creates acronym from first letters. Example: "Central Processing Unit" → "CPU".

Program 128: Expand Acronym Write a program that expands given acronym to full form using provided dictionary.

Program 129: Check if Valid Identifier Write a program that checks if string is valid identifier (starts with letter/underscore, contains only alphanumeric/underscore).

Program 130: Count Substring Overlapping Write a program that counts overlapping occurrences of substring. Example: "aaa" in "aaaa" → 2 times.